

UNIT ONE

LINEAR EQUATIONS, FUNCTIONS AND THEIR APPLICATIONS

INTRODUCTION

Profitability is the major reason for owners and management in operating a business. Effective management decisions must be made which affects produce mix, total revenue, total cost to meet the desired level profitability and linear mathematical models like Break Even Analysis are important tools to facilitate management decisions.

Linear mathematics is significant for a number of reasons. First, many of the real world phenomenon, which we might be interested in representing mathematically, either are linear or can be approximated reasonably well using linear relationships. As a result, linear mathematics is widely applicable. Second, the analysis of linear relationships is generally easier than that of non-linear relationships. Lastly, the methods of analyzing non-linear relationships are sometimes similar to, or extension of, those used in linear mathematics. Consequently, a good understanding of linear mathematics is prerequisite to study non linear mathematics. Hence, it should not be underestimated.

Linear Equations and their characteristics

Linear equations are equations whose term (the parts separated by plus, minus, and equal signs) are constant, or a constant times one variable to the first power. From the above definition, we can characterize linear equations as:

a) Linear equations are first degree equations.

Each variable in the equation is raised (implicitly) to the first power. The presence of terms having exponents other than 1 would exclude an equation from being considered linear.

Example 1.1

$x^2 + y + 3x = 16$ is not linear because of the presence of the second power term, x^2 .

Other example:

$$\left. \begin{array}{l} \sqrt{x+9} = Y \\ x^2 + y = x \end{array} \right\} \dots (x+9)^{1/2} \text{ are not linear}$$

b) Linear equations involve a product of a constant term with a variable. The presence of terms involving a product of variables would exclude an equation from being considered linear.

Example 1.2

$$\left. \begin{array}{l} 2x - 3y = 7 \\ 2y - x = 4 \\ x/2 + 1/4 y = 4 \end{array} \right\} \text{ all are linear because all are constant times the product of two variables.}$$

But, $2x + 3xy = 7$ is not linear, because $3xy$ is a constant times the product of two variables.

Generally, a linear equation involving two variable x and y has the standard form $ax + by = C$, where a , b and c are constants and a and b cannot both equal Zero and a linear equation involving n variables $x_1, x_2, x_3, \dots, x_n$ has the general formula $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$, where $a_1, a_2, \dots, a_n$ and b are constants and $\neq$ (different from) Zero.

Note: The names of variables may be different form x and y like a , b , u and others. But customarily we use x and y .

The form of an equation may not always be obvious. Initially, the equation

$$2x = \underline{5x - 2y} + 10$$

might not appear to be linear. However, multiplying both sides of the equation by 4 and moving all variables to the left side yields:

$$4(2x = 5x - 2y + 10)$$

$$4$$

$$\Rightarrow 8x = 5x - 2y + 40$$

$$\Rightarrow 3x + 2y = 40, \text{ which is a linear equation.}$$

Solutions and Forms of Linear Equations

Solutions of Linear Equations

Given a linear equation having the form $ax + by = c$, the solution set for the equation *is the set of all order at pairs* (x, y) , which satisfy the equation. Using set notation, the solution set S can be specified as:

$$S = \left\{ (x, y) / ax + by = C \right\}$$

Verbally, this set notation states that the solution set S consists of elements (x, y) such that the equation $ax + by = C$ is satisfied. Stated differently that S consists of all ordered pairs (x, y) such that $ax + by = C$. Note: For any linear equation, S consists of an infinite number of elements: that is, there is an infinite number of pairs of values (X, Y) which satisfy a linear equation having the form $ax + by = C$. To determine any pair of values which satisfy a linear equation, assume a value for one of the variables, substitute this value in to the equation, and solve for the corresponding value of the variable.

Example 1.3

Consider a linear equation $y = 2x + 500$. Determine:

- sets of numbers which will satisfy the equation?
- the pair of values which satisfies the equation when $x = 20$
- the pair values which satisfies the equation when $y = 0$

 **Note:** When we write the ordered pairs, the value of x is written first and that of y is written next after comma.

Application: A company manufactures two different products. For the coming week 120 hours of labor are available for manufacturing two products. Work hours can be allocated for production of either product. In addition, since both products generate a good profit, management is interested in using all 120 hours during the week. Each unit produced of product A requires 3 hours of labor and each unit of product B requires 2.5 hours.

Required:

- Define an equation which states that total work-hours used for producing the two units?
- How many units of product A can be produced if 30 units of B are produced?
- If management decides to produce one product only, what is the maximum quantity which can be produced of product A?
- If the management decides to produce product B only, what is the maximum quantity of product B to be produce?

Slope of Linear Equations

As we have already discussed in the previous section that, linear function, $ax + by = C$ where a , b , and c are non-zero constant numbers is very important function in the real world business application problems. The solution set of an equation in two variables is the set of all solutions of the equation.

The graph of $ax + by = c$ is a straight line and the slope of the line through two points (x_1, y_1) and (x_2, y_2) is given by:

$$\text{Slope} = m = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_2 \neq x_1.$$

Slope of straight line or line segment is the ratio of rise or fall to the run between two points on the line, where rise or fall is a vertical segment and run is a horizontal separation of two points.

$$\text{Hence, Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{\text{Difference of y's}}{\text{Difference of x's}} = \frac{y_2 - y_1}{x_2 - x_1}$$

Any straight line, with the exception of vertical lines can be characterized by **slope** - *refers to the inclination of a line* - whether it rises or falls as we move from left to right along the x axis - and the rate at which the line rises or falls (in other words, how steep the line is).

The slope of a line may be positive, negative, Zero, or undefined

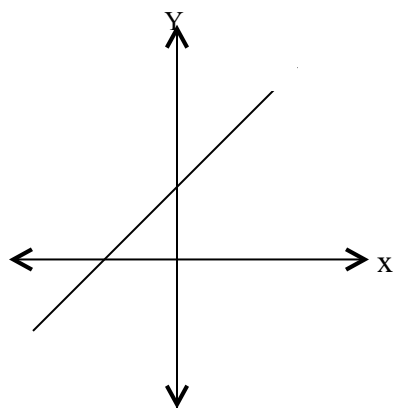
Positive Slope - A line with a positive slope rises from left to right, or runs up hill. For such a line the value of y increases as x increases (conversely, y decreases as x decreases).

Negative Slope- A line having a negative slope falls from left to right, or runs down hill. For such a line the value of y decreases as x increases (or conversely, y increases as x decreases), hence, x and y behave in an inverse manner.

Zero Slope - A line having a zero slope is horizontal. As x increases or decreases y stays constant. This is the case where the value of y is the same in pairs of values

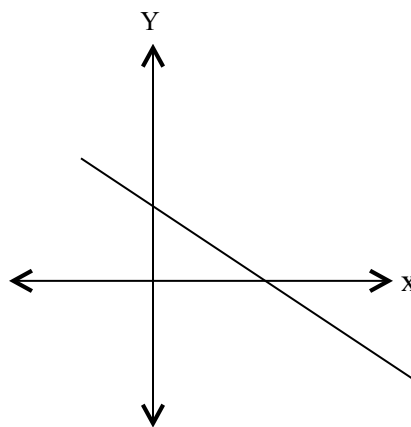
Undefined Slope- Vertical lines (or the form $x = k$) have a slope which is undefined. Since x is constant, we cannot observe the behavior of y as x changes.

As x increases, y increases



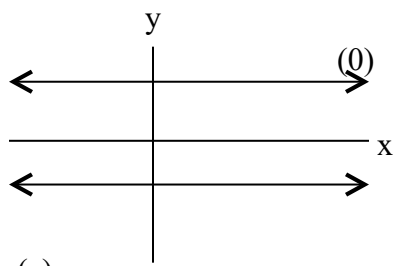
(a) Positive slope

As x increases, y decreases



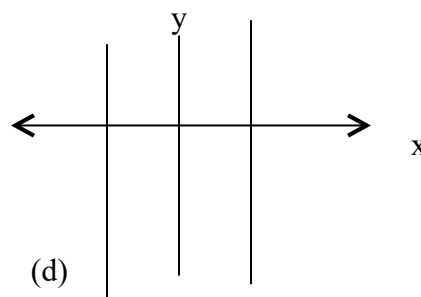
(b) Negative slope

As x increases or decreases, y remains constant, ($y=k$)



(c) Zero slope

x is constant regardless of value of y. ($x=k$)



(d) Slope undefined

 **Note:** The sign of the slope (number) indicates whether the line is rising or falling. The magnitude (absolute value or the number) of the slope indicates the relative steepness of the line. The larger the absolute value of the slope, the steeper the angle at which the line rises or falls.

Example 1.4 : Find the slope of the line through the pair of points (-1, 4) and (1, 8) and interpret it.

Forms of Linear Equations

Generally, there are three forms of linear equations which are considered as important tools in linear mathematical models: *slope - intercept form, point - slope form and Two - Points form.*

a) Slope - Intercept form

If the equation of a line is written in the form of $y = mx + b$, it is called slope - intercept form, where b is y intercept and m is the slope.

Example 1.6

Write equation of a line with slope of 2 and y intercept of 5.

Solution

Equation of a straight line where slope and y - intercept are given is defined as:

$$y = mx + b$$

Here, $m = 2$ and $y = 5$. Then by substituting these numbers to the equation, we will get equation of a line:

$$y = 2x + 5.$$

Example: Write equation of a line with slope = $\frac{1}{2}$ and y intercept is $\frac{2}{5}$?

b) Point - slope Form

The point slope form is extremely important, since it enables us to find linear mathematical models for equations of straight lines if we know its slope and coordinate of a point on the line. To develop a point-slope form of linear equation, slope should be known. A linear equation of a line having slope m and which passes through the point (x_1, y_1) is given by:

$$m = \frac{y - y_1}{x - x_1} \Rightarrow y - y_1 = m(x - x_1) \text{ is called Point-slope form of linear equation}$$

Example 1.6

If a straight line passes through points $(\frac{1}{2}, 5)$ with the slope of 4, what will be equation of a line?

c) Two points form

The slope m of the straight line connecting two points (x_1, y_1) and (x_2, y_2) is

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

When this is expanded, it takes a form of:

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \text{ and is called two - point slope form of linear equations.}$$

Example 1.7: What will be equation of a line that passes through points (2, 4) and (5, 12)?

1.2.4. Interpreting the slope and Y intercept

In many applications of linear equations, the slope and y intercept have meaningful interpretations. As slope is the change in the value of y if x increases by 1 unit, intercept is the point at which the line crosses either of the axis.

X - Intercept is the point where the line crosses X axis. It is computed by setting the value of Y zero (i.e. $Y = 0$).

Y - Intercept is the point where the graph of the equation crosses the Y - axis. The Y intercept represents the ordered pairs found by setting $x = 0$

Example 1.7

Let Y be revenue and X be advertisement. If $\frac{1}{2}y = 2x + 500$, compute and verbally interpret the meaning of slope of a line? Determine x and y intercept?

Application. The chairman is trying to forecast the number of tourists who would be visiting the city of Bihar Dar in the coming years. Some students from the local university have examined the problem as a part of their class project. They estimated the current year's Volume as 60,000 tourists. They have also projected that the volume in the next year will be 67,500 and that the volume in the future will continue to grow at the same rate. Determine equation of a line which relates the number of tourists, p to Years t as measured from this current year?

1.3 Graphical Characteristics

Graphing Two - Variable equations

A linear equation involving two variables graphs as a straight line in two dimensions.

Steps to graph this type of linear equation:

- i) identify and plot the coordinates of any two points which are on the line.
- ii) Connect the two points with a straight line
- iii) extend the straight line in both sides as desirable as for your purpose

Note: The coordinates of two points are found by identifying any two members of the solution. The easiest points to identify are those found by setting one variable equal to Zero and solving for the other variable.

Example 1.8

Graph the equation $2x + 4y = 16$

 **Note:** Any two variables linear equation having the form $ax + by = 0$ graphs as straight line which passes through the origin. The unique property of this equation is that the right side, c. equals zero

? Graph a linear equation of $4x - 7y = 0$

A Linear Functions and Their applications

In this sub-section we extend the material presented in the previous sub section by presenting a discussion of linear functions.

Linear Functions

A linear function, f, involving one independent variable x and a dependent variable y has the general form:

$$Y = f(x) = a_1x + a_0, \text{ where } a_1 \text{ and } a_0 \text{ are constants, } a_1 \neq 0.$$

For a linear function having the form, $f(x) = a_1x + a_0$, a change in the value of y is directly proportional to a change in the value of x. This rate of change is constant and represented by the slope a_1 .

Example 1.8.

Imagine that you have taken a job as a sales person. Your employer has stated that your salary will depend upon the number of units you sell each week.

If we let y = weekly salary in birr

x = number of units sold each week

The dependency stated by your employer can be represented by the equation.

$y = f(x)$, where f is the name of the salary function.

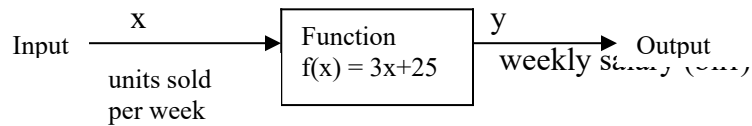
Suppose your employer has given you the following equation for determining your weekly salary:

$$y = f(x) = 3x + 25$$

If you sold 100 units in the week, your weekly salary will be:

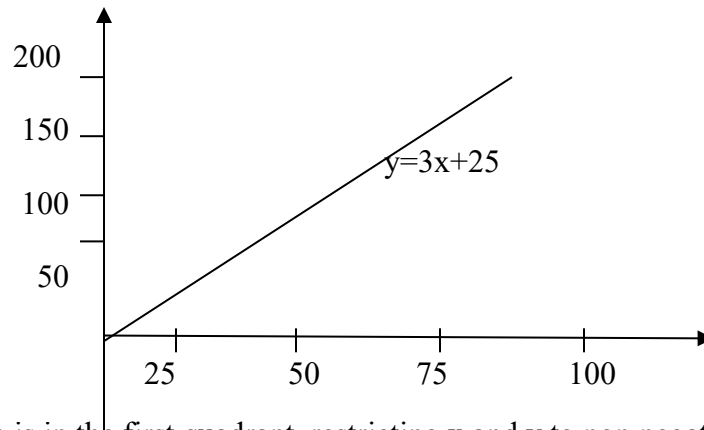
$$y = 3(100) + 25$$

$$y = 325$$



? What will be your weekly salary if you sell 75 units?

In the above example, the change in your weekly salary is directly proportional to the change in the number of units sold. That is, the slope 3 indicates the increase in weekly salary associated with each additional unit sold.



Note: The graph is in the first quadrant, restricting x and y to non negative values. Does this make sense? (Give your opinion)

A linear function, f , involving two independent variables x_1 and x_2 and a dependent variable Y has the general form:

$$y = f(x_1, x_2) = a_1 x_1 + a_2 x_2 + a_0, \text{ where}$$

a_1 and a_2 are (non Zero) constant and a_0 is a constant.

Here, Variable y depends jointly on the values of x_1 and x_2 . The value of y varies in direct proportion to changes in the values x_1 and x_2 . Specifically, if x_1 increases by 1 unit, y will change by a_1 units.

Example 1.9

Assume that a sales person's salary depends on the number of units sold of each of two products which is given by:

$$y = 5x_1 + 3x_2 + 25, \text{ where, } x_1 = \text{number of products sold}$$

$$x_2 = \text{number of product 2}$$

$$y = \text{Weekly salary}$$

This salary function suggests a base weekly salary of Birr 25 and commissions per unit sold of Birr 5 and 3, respectively, for Product 1 and 2

- Suppose the commission is 2% and 1.5% respectively for product 1 and 2. The weekly sales value for product 1 is Birr 10,000 and Birr 9000 for product 2. What will be the weekly salary of the salesperson?
- Suppose also that the total units sold per week are 25 and 50 respectively, What will be the weekly salary of the salesperson?
- If weekly salary is Birr 400, and the weekly sales are 10,000 each and the commission on product 1 is 1%, what will be the commission on product 2?

A linear function involving n independent variables $x_1, x_2, x_3, \dots, x_n$ and a dependent variable y has the general form:

$y = f(x_1, x_2, \dots, x_n)$ or $y = a_1x_1 + a_2x_2 + \dots + a_nx_n + a_0$, where $a_1, a_2, \dots, a_n$ are (non-zero) constants and a_0 is a constant.

Linear cost Function

Organizations are concerned with costs, because they reflect Birrs flowing out of the organization. The out flows usually could be pay for salaries, raw materials, supplies, and so forth. Therefore, the cost function, $C(x)$ is defined by $C(x) = mx + b$, where m is variable cost per unit, b is fixed cost and x is the volume of output.

Total cost has two components: Total Variable cost and total fixed cost. These two costs can be added to determine total cost.

Example 1.10

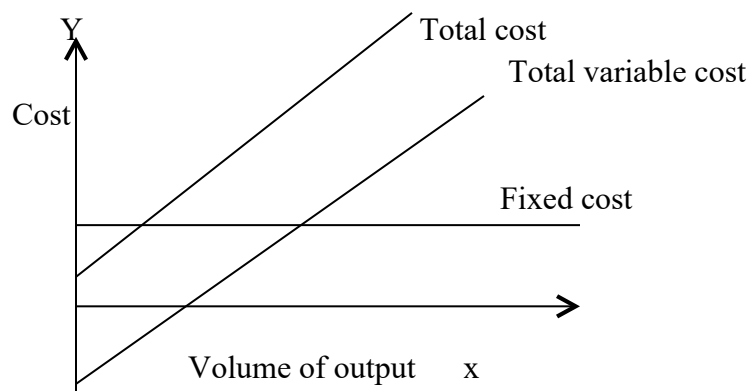
The cost function,

$C(x) = 0.4x + 18,000$ had variable costs which vary with the units produced and a fixed cost of 18,000.

- Total variable costs vary with the level of output and computed as the product of variable cost per unit of output and the level of output. Hence,

$$\text{Total variable cost (TVC)} = \text{Variable cost/unit} \times \text{Units produced},$$

$$\text{TVC} = \text{Vc/Unit} \times \text{Units produced}.$$
- Total Fixed cost remains constant regardless of the change in the level of joints for certain predetermined level of output.



? why do you think the graph of total cost function start from the fixed cost?

👉 **Note:** Total Variable cost starts from origin since at Zero level of unit produced, no variable cost will be incurred. But, fixed cost will not start from origin because regardless of the level of units produced the organization incurs this cost.

Application: A firm which produces a single product is interested in determining the function that express annual total cost y as a function of the number of units produced x . Accountants indicate that fixed costs each year are birr 50,000. They also have estimated that raw material costs for each unit produced are Birr 5.5, and labor costs per unit are Birr 1.50 in the assembly department, Birr 0.75 in the finishing room, and Birr 1.25 in the packaging and shipping department.

- a) Determine the variable cost per unit and cost function
- b) What will be the total variable cost if 20,000 units are produce?
- c) What is the marginal cost?
- d) What will be the total cost, if 10,000 units are produced?

? If the total factory cost, y , of making x units of a product is $y = 10x + 500$ and if 5000 Units are made, what is

- i) total Variable cost
- ii) Total Cost
- iii) Variable cost per unit
- iv) Marginal cost of last unit made?

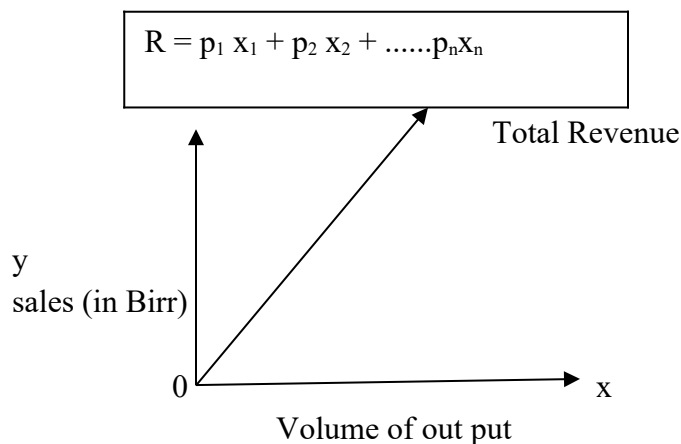
Linear Revenue Function

The money which flows in to an organization from either selling products or providing services is often referred to as **revenue** and is given by:

Total Revenue = (Price) (Quantity sold or service rendered)

The function $R(x) = mx$, where m is the rate of selling price per unit item and x is the total volume of output, is called the **Revenue function**.

An assumption in this relationship is that the selling price is the same for all units sold. Assume a firm sells n products. If x_i equals the number of units sold of product i and p_i equals the price of product i , the function which allows you to compute total revenue from the sale of n products is:



👉 **Note:** the revenue function starts from origin. This is because when no units are sold regardless of the amount of price, the total revenue generated will be zero.

Example 1.11

A firm is selling three products x_1 , x_2 and x_3 . If it sells 20 units of x_1 , 15 units of x_2 and 25 units of x_3 at a price of Birr 100, 125 and 350 respectively,

- a) what will be the total revenue?
- b) If price of x_2 is increased by 20%, what will be the new total revenue?
- c) What will be the new total revenue if the price of x_1 is reduced by 10% and the number of units of x_3 sold is increased by 5%?

Linear Profit function

Profit for an organization is the difference between total revenue and total cost stated in equation form:

$$\text{Profit} = \text{Total Revenue} - \text{Total Cost}$$

If total revenue = $R(x)$ and total cost = $C(x)$, where x equals the quantity produced and sold, then profit is defined as:

$P(x) = R(x) - C(x)$ and there are three conditions:

- i) Total revenue exceeds total cost; i.e. profit is positive and is referred to as net gain, or net profit.
 $R(x) > C(x)$

- ii) Total cost exceeds total revenue, profit is negative, and is referred to as a net loss or deficit.

$$C(x) > R(x)$$

- iii) Total cost equals total revenue, i.e. no profit, no loss

$$R(x) = C(x)$$

Note: when the revenue and cost are linear functions of the same variable(s), the profit function is a linear function of the same variable (s). Moreover, the additional Birr generated as the result of producing and selling one more unit is called Marginal profit and is used as slope of profit function.

Example 1.12

A firm sells a single product for Birr 65 per unit. Variable costs per unit are Birr 20 for materials and Birr 27.50 for labor. Annual fixed costs are Birr 100,000.

- i) Construct the profit function stated in terms of x
- ii) What profit is earned if annual sales are 20,000?
- iii) What is Marginal profit of 20th unit?

Break Even Analysis (BEA)

In this section, we will discuss about Break Even Models, a set of planning tools which can be, and has been, very useful in managing organizations. It is defined as a technique for studying the relationship between cost, volume and profit and enables to understand the impact of changes in the level of output on cost and profit. Hence, it is called V-C-V Analysis and Break even analysis.

Break Even Analysis for manufacturing firms

Break even analysis focuses up on the profitability of a firm. Of specific concern in breakeven analysis is identifying the level of operation or level of output that would result in a zero profit. This level of operation or output is called breakeven point. The breakeven point is a useful reference point in the sense that it represents the level of output at which total revenue equals total revenue, i.e. net profit is zero. Any change from this level of operation results in either profit or loss.

Important Assumptions:

- Both total revenue and total cost function are linear.
- Total variable costs change directly and proportionately with the volume of output. But, variable costs per unit are constant or assumed to be constant regardless of the level of output.
- Fixed costs remain unchanged irrespective of the determined level of output (at least in the short term).
- Selling price per unit is constant. Where the selling price is not constant, average price is sometimes chosen for purposes of conducting the analysis.
- Price per unit is greater than variable cost per unit. If price per unit is less than the cost per unit, a firm will lose money on every unit produced and sold. Break even conditions never exist.

Profit cannot be realized at all level of output even if the selling price is greater than production cost. In other words, there is no guarantee to make profit for the mere fact that the selling price is greater than the variable cost per unit. To determine the level of output which generates profit, we should have to know the breakeven point, which is primary objective of BEA.

Procedures in determining breakeven point

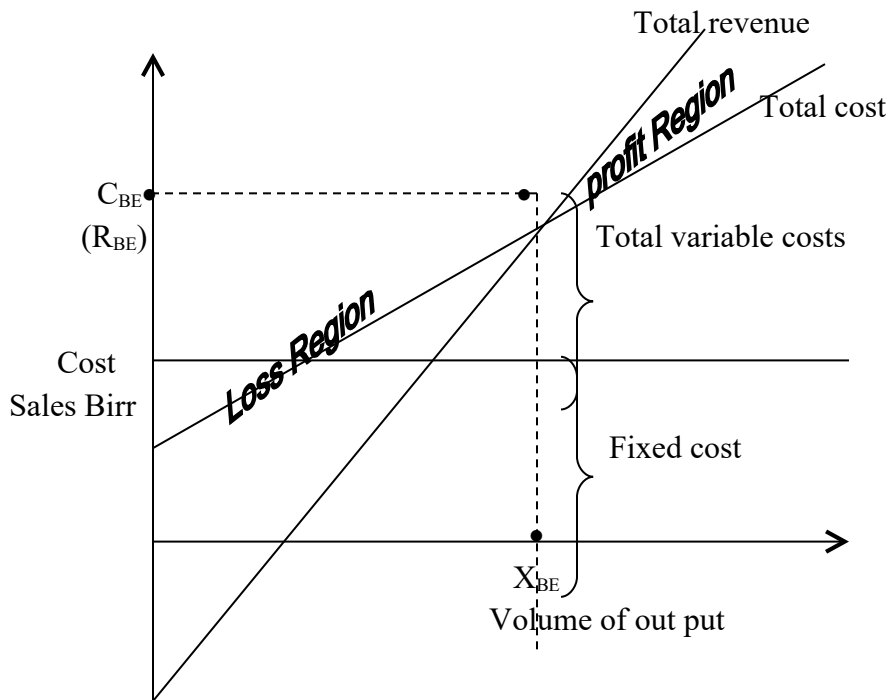
- i. Formulate total cost as a function of x, the level of output.
- ii. Formulate total revenue as a function of x
- iii. Set $C(x)$ equal to $R(x)$ and solve for x. The resulting value of x is breakeven level of output and might be denoted by X_{BE} .

An alternative to step iii is to construct a profit function,

$P(x) = R(x) - C(x)$, set x equals to zero and solve for x .

Graphic Representation

The graph of the cost behavior and the revenue behavior on the same X-Y plane is called the Break-even Chart and is shown graphically as:



Explanations

- Costs and revenues (in Birr) are depicted on vertical axis (y-axis) and volume of output is depicted on horizontal (x-axis)
- Fixed costs line is drawn horizontally parallel to x-axis representing it is unchanging for any level of output.
- Total revenue is drawn as straight line starting from origin (zero) and total cost line is drawn from the vertical axis where the fixed cost starts.
- Total revenue and total cost lines intersect (will be equal) at a point called Breakeven point. This point can be expressed as X_{BE} , in terms of units or R_{BE} (or C_{BE}) representing break even revenue or cost determined in Birr.
- Any level of output less than X_{BE} results in loss for an organization and any level of output greater than X_{BE} results in profit or surplus.
- The point where the line drawn perpendicular from the breakeven point intersects the horizontal axis (X- axis) indicates break even volume in units (X_{BE}) or as a percent of capacity.
- The point where the line drawn perpendicular from breakeven point intersects y-axis indicates break even volume in dollars or breakeven cost.
- The area between horizontal and the fixed line represents fixed cost line and the area between fixed cost line and total cost line represents total variable cost in dollars for different level of output.

Mathematically, the breakeven point may be expressed in terms of:

- 1) Volume of output (level of activity)

$P(x) = R(x) - C(x)$, but at breakeven point P is zero, P is profit.

$$\Rightarrow R(x) = C(x)$$

$$\Rightarrow px = TVC + Fc$$

$$\Rightarrow px = (vc/unit)(x) + Fc, \text{ x is the level of out put}$$

Tvc is total variable cost
Fc is total fixed cost

$$\Rightarrow px - (vc/unit)(x) = Fc$$

$$\Rightarrow x(p - vc/unit) = Fc$$

$$\Rightarrow x = \frac{Fc}{(p - vc/unit)}, \text{ where x is the breakeven level of unit}$$

$$\text{Hence, } x_{BE} = \frac{Fc}{(p - vc/unit)}$$

Note: p-vc/unit is profit margin, which is also called contribution to fixed costs and profit. Only when total profit contribution exceeds fixed costs will a positive profit exist.

2) Total Birr sales

Break-even can also be expressed in Birr value as:

$$R(x) = C(x)$$

$$px = Tvc + Fc$$

$$px = (vc/unit)(x) + Fc$$

$$px - (vc/unit)(x) = Fc$$

$$x(p - vc/unit) = Fc$$

Since our interest is to express breakeven point in terms of revenue (Birr value), we should make p free. Hence,

$$\Rightarrow x(p - vc/unit) = Fc \dots \text{divide both sides by x, which results in}$$

$$\Rightarrow p - vc/unit = \frac{Fc}{x} \dots \text{divide both sides by p which results in}$$

$$\frac{p - vc/unit}{p} = \frac{Fc}{px}, \text{ since px is revenue which is expressed in Birr}$$

$$\Rightarrow px = \frac{Fc}{(p - vc/unit/p)}$$

$$\Rightarrow px = \frac{Fc}{1 - vc/unit} \dots \text{formula to determine breakeven point in sales value.}$$

or since $\frac{p - vc/unit}{p}$ is contribution margin ratio (CRM), we can rewrite the formula as:

$$R_{BE} = px = \frac{Fc}{CRM}$$

$$\text{Or } R_{BE} = px = \frac{Fc(p)}{p - vc/unit} = \frac{Fc(p)}{CM}, \text{ CM is contribution margin given by } p - vc/unit$$

3. Percentage of production or sales capacity

Breakeven point as a percentage of production or sales = $\frac{\text{Break even units or sales}}{\text{Capacity of production or sales}}$

Example 1.12

A group of engineers are interested in forming a company to produce smoke detectors. They have gone a design stage and estimate that variable costs per unit including materials, labor and marketing costs are Birr 22.5. Fixed costs associated with formation, operation, etc totals Birr 250,000. they estimate that the selling price will be Birr 30 per detector.

- Determine the number of smoke detectors which will be sold in order for the firm to break even on the venture?
- A market study indicates that the firm can expect to sale approximately 30,000 detectors at Birr 30. Determine the expected level of profit at this level of output?
- Determine Break even volume in Birr and variable costs at Breakeven point
- Represent graphically
- If the firm contemplates to generate a profit of Birr 30,000, what should be the level of output to be produced and sold?

Break even Analysis for merchandising firms

Dear learners, in our previous discussion we tried to cover break even analysis for those firms which produce and sell their products. In the real world, this is not always the case. This is because there are firms which do not produce products by themselves; but purchase from other organizations and hold for some time and sell it to others.

For those firms that purchase products and sell them at a price that is presumably above the cost, the break even analysis is treated differently.

Suppose, the cost of an item is Birr 130 and priced to be sold at Birr 200. The mark up is equals to the retail price minus cost. Hence, Mark up = Retail price – Cost.

Since margin is mainly used in the financial statements of such business organizations, it is defined as a markup percentage of retail price.

$$\text{Hence, Margin} = \frac{\text{Mark up}}{\text{Retail price}} = \frac{\text{Retail price} - \text{Cost}}{\text{Retail price}}$$

From this, the mark up is Birr 70 (200-130) and the margin is 35 % (70/200). This means that 35% of retail price of Birr 200 is margin and the other 65% of the retail price is cost. Thus, cost of goods sold (COGS) equals 65% of the retail price or revenue. If we denote x as the sales volume in Birr, COGS= 0.65x

Example 1.13

If a merchandising company incurs selling expense of 10%, and FC of Birr 12,000, what will be the total cost? At what level of sales that the company break-even?

If the relationship between cost and sales is written in the general slope intercept form,

$$y = mx + b, \text{ where } y \text{ is total cost and } x \text{ is total sales volume in Birr}$$

At breakeven point, total cost equals to total sales, $y = x$

$$mx + b = x, \text{ m is variable cost per Birr of sales and b is fixed cost.}$$

$$\Rightarrow x - mx = b$$

$$\Rightarrow x(1 - m) = b$$

$$\Rightarrow x = \frac{b}{(1 - m)}$$

$$\Rightarrow x = x_{BE} = \frac{b}{(1 - m)} = \frac{\text{Fixed cost}}{1 - \text{vc/Birr of sales}} \dots \text{Break even formula}$$

Exercise

ABC merchandising firm plans to work on a margin of 46% of retails and it incurs other variable cost of Birr 0.16 per Birr of sales. A fixed cost equals to Birr 25,000.

- Find the equation relating total cost to sales?
- Find the BEP?
- What would be net profit before taxes if sales are Birr 100,000?
- If the firm wants to make profit before taxes of Birr 20,000, what should be the revenue level?